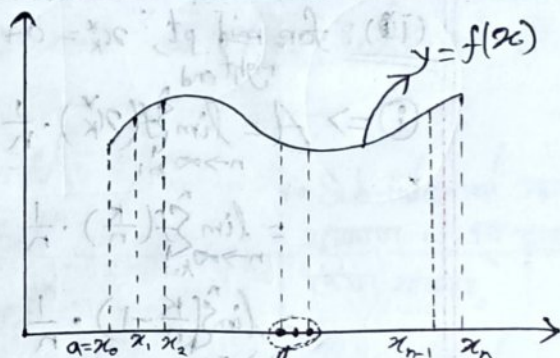


Riemann Sum :

Let,

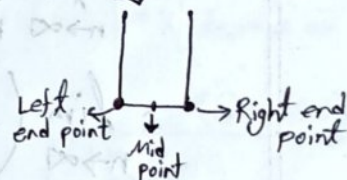
$y = f(x)$ be a function on the closed interval $[a, b]$



(i) Divide the interval $[a, b]$ into n -subintervals

$[x_{k-1}, x_k]$ of width $\Delta x_k = x_k - x_{k-1}$

where, $a = x_0 < x_1 < x_2 < \dots < x_n = b$



(ii) Choose a number x_k^* in each subintervals $[x_k, x_{k-1}]$

these n -numbers $x_1^*, x_2^*, \dots, x_n^*$ are called sample points.

(iii) Calculate the value $f(x)$ at $x = x_k^*$, then the Riemann

Sum:

$$* A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k *$$

Since, the sample points can be chosen arbitrarily, some common choices are given by -

(i) Left end point: $x_k^* = a + (k-1)\Delta x$;

(ii) Right end point: $x_k^* = a + k\Delta x$;

(iii) Mid point: $x_k^* = a + (k - \frac{1}{2})\Delta x$; where $\Delta x = \frac{b-a}{n}$

* Ex-1: Find the Riemann Sum of $f(x) = x - 1$ at the interval $[0, 1]$

using $\Sigma = (n+1)^2 - 1$ for sum of first n natural numbers

- (i) Left end point ; Solⁿ: $-\frac{1}{2}$
 (ii) Right end point ; Solⁿ: $-\frac{1}{2}$
 (iii) Mid point ; Solⁿ: $-\frac{1}{2}$

• Sub-interval বলা থাকলে n এর মান দেয়া থাকবে,

Solⁿ:

(i) $A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$ — (i)

$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$

$x_k^* = a + (k-1) \Delta x$
 $= 0 + (k-1) \frac{1}{n}$

$\therefore x_k^* = \left(\frac{k-1}{n}\right)$

(i) $\Rightarrow A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \frac{1}{n}$
 $= \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{k-1}{n}\right) \frac{1}{n}$

$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k-1}{n} - 1 \right] \frac{1}{n}$

$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k-1}{n^2} - \frac{1}{n} \right]$

$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k}{n^2} - \frac{1}{n^2} - \frac{1}{n} \right]$

$= \lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \frac{k}{n^2} - \sum_{k=1}^n \frac{1}{n^2} - \sum_{k=1}^n \frac{1}{n} \right]$

$f(x) = x - 1$
 $f\left(\frac{k-1}{n}\right) = \frac{k-1}{n} - 1$

(iii)
 $\sum_{k=1}^n k = 1+2+3+\dots+n = \frac{n(n+1)}{2}$
 $\sum_{k=1}^n 1 = 1+1+1+\dots+n = n$

• 'k' depends on 'n'
 (Series)

$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sum_{k=1}^n k - \frac{1}{n^2} \sum_{k=1}^n 1 - \frac{1}{n} \sum_{k=1}^n 1 \right]$

$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \cdot \frac{n(n+1)}{2} - \frac{1}{n^2} \cdot n - \frac{1}{n} \cdot n \right]$

$= \lim_{n \rightarrow \infty} \left[\frac{n+1}{2n} - \frac{1}{n} - 1 \right]$

$= \lim_{n \rightarrow \infty} \left[\frac{n}{2n} + \frac{1}{2n} - \frac{1}{n} - 1 \right]$

$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} + \frac{1}{2n} - \frac{1}{n} - 1 \right]$

$= \frac{1}{2} + 0 - 0 - 1$

$= -\frac{1}{2}$
 $\therefore A = -\frac{1}{2}$

First, check the right page, then come back here →

(ii): for mid pt, $x_k^* = a + K\Delta x = 0 + K \cdot \frac{1}{n} = \frac{K}{n}$

$$\textcircled{i} \Rightarrow A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \cdot \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{K}{n}\right) \cdot \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{K}{n} - 1\right) \cdot \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{K}{n^2} - \frac{1}{n}\right)$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{K}{n^2} - \sum_{k=1}^n \frac{1}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \cdot \sum_{k=1}^n K - \frac{1}{n} \sum_{k=1}^n 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \cdot \frac{n(n-1)}{2} - \frac{1}{n} \cdot n \right]$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n-1}{2n} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{2n} - \frac{1}{2n} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} \left(-\frac{1}{2} - \frac{1}{2n} \right)$$

$$= -\frac{1}{2} - 0$$

$$= -\frac{1}{2} \text{ (Ans.)}$$

$$\left| \begin{array}{l} f(x) = x - 1 \\ f\left(\frac{K}{n}\right) = \frac{K}{n} - 1 \end{array} \right| \quad \left| \begin{array}{l} f\left(\frac{2K-1}{2n}\right) = \frac{2K-1}{2n} - 1 \end{array} \right|$$

(iii) for mid pt, $x_k^* = a + (K - \frac{1}{2})\Delta x = 0 + (K - \frac{1}{2}) \cdot \frac{1}{n} = \frac{2K-1}{2n}$

$$\textcircled{i} \Rightarrow A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \cdot \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{2K-1}{2n}\right) \cdot \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{2K-1}{2n} - 1 \right] \cdot \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{2K-1}{2n^2} - \frac{1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{2K}{2n^2} - \frac{1}{2n^2} - \frac{1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \frac{K}{n^2} - \sum_{k=1}^n \frac{1}{2n^2} - \sum_{k=1}^n \frac{1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sum_{k=1}^n K - \frac{1}{2n^2} \sum_{k=1}^n 1 - \frac{1}{n} \sum_{k=1}^n 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \cdot \frac{n(n-1)}{2} - \frac{1}{2n^2} \cdot n - \frac{1}{n} \cdot n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n-1}{2n} - \frac{1}{2n} - 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n}{2n} - \frac{1}{2n} - \frac{1}{2n} - 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[-\frac{1}{2} - \frac{1}{n} \right] = -\frac{1}{2} - 0 = -\frac{1}{2} \text{ (Ans.)}$$

Ex-2:

Find the Riemann Sum of $f(x) = 3x + 1$ over the interval $[2, 6]$ with 4 subintervals using:-

- (i) Left end point ; Ans $\rightarrow 46$
 (ii) Right end point ; Ans $\rightarrow 58$
 (iii) Mid point ; Ans $\rightarrow 52$

$n = 4$

Solⁿ:

$$(iii) A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

$$= \sum_{k=1}^4 f(x_k^*) \Delta x$$

$$= f(x_1^*) \Delta x + f(x_2^*) \Delta x + f(x_3^*) \Delta x + f(x_4^*) \Delta x \quad \text{--- (i)}$$

$$\Delta x = \frac{b-a}{n} = \frac{6-2}{4} = 1$$

For mid point,

$$x_k^* = a + (k - \frac{1}{2}) \Delta x$$

$$= 2 + (k - \frac{1}{2}) \cdot 1$$

$$= 2 + k - \frac{1}{2} = k + \frac{3}{2}$$

$$x_1^* = 2 + \frac{3}{2} = \frac{7}{2} ; f(x_1^*) = f(\frac{7}{2}) = 3 \cdot \frac{7}{2} + 1 = \frac{23}{2}$$

$$x_2^* = 2 + \frac{5}{2} = \frac{9}{2} ; f(x_2^*) = f(\frac{9}{2}) = 3 \cdot \frac{9}{2} + 1 = \frac{28}{2}$$

$$x_3^* = 2 + \frac{7}{2} = \frac{11}{2} ; f(x_3^*) = f(\frac{11}{2}) = 3 \cdot \frac{11}{2} + 1 = \frac{34}{2}$$

$$x_4^* = 2 + \frac{9}{2} = \frac{13}{2} ; f(x_4^*) = f(\frac{13}{2}) = 3 \cdot \frac{13}{2} + 1 = \frac{40}{2}$$

Now, (i) $\Rightarrow$

$$A = f(x_1^*) \Delta x + f(x_2^*) \Delta x + f(x_3^*) \Delta x + f(x_4^*) \Delta x \quad | \quad \Delta x = 1$$
$$= \frac{17}{2} + \frac{23}{2} + \frac{29}{2} + \frac{35}{2}$$
$$= 52 \text{ (Ans.)}$$

(ii) For right end point,

$$x_k^* = a + k \Delta x = 2 + k \cdot 1 = 2 + k = k + 2$$

$$x_1^* = 1 + 2 = 3 ; f(x_1^*) = f(3) = 3 \cdot 3 + 1 = 10$$

$$x_2^* = 2 + 2 = 4 ; f(x_2^*) = f(4) = 4 \cdot 3 + 1 = 13$$

$$x_3^* = 3 + 2 = 5 ; f(x_3^*) = f(5) = 3 \cdot 5 + 1 = 16$$

$$x_4^* = 4 + 2 = 6 ; f(x_4^*) = f(6) = 3 \cdot 6 + 1 = 19$$

$$\text{Now, (i)} \Rightarrow A = 10 \times 1 + 13 \times 1 + 16 \times 1 + 19 \times 1 = 58 \text{ (Ans.)}$$

(iii) For left end point,

$$x_k^* = a + (k-1) \Delta x = 2 + (k-1) \cdot 1 = k + 1$$

$$x_1^* = 1 + 1 = 2 ; f(x_1^*) = f(2) = 3 \cdot 2 + 1 = 7$$

$$x_2^* = 2 + 1 = 3 ; f(x_2^*) = f(3) = 3 \cdot 3 + 1 = 10$$

$$x_3^* = 3 + 1 = 4 ; f(x_3^*) = f(4) = 3 \cdot 4 + 1 = 13$$

$$x_4^* = 4 + 1 = 5 ; f(x_4^*) = f(5) = 3 \cdot 5 + 1 = 16$$

$$\text{Now, (i)} \Rightarrow A = 7 + 10 + 13 + 16 = 46 \text{ (Ans.)}$$

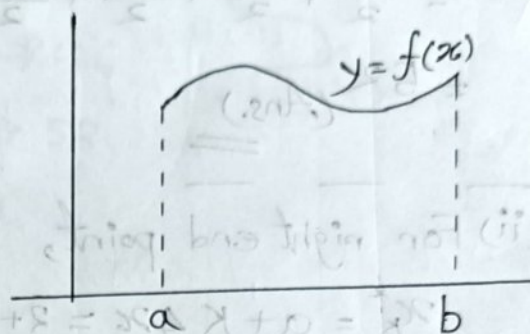
Area Under the Curve

• Net sign area/area:

Let the net sign area of $f(x)$ over the interval $[a, b]$ is

$$A = \int_a^b f(x) dx$$

where, a is lower limit
 b is upper limit



Formula :

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$2. \int \frac{dx}{x} = \ln|x| + C$$

$$3. \int dx = x + C$$

$$4. \int e^x dx = e^x + C$$

$$5. \int e^{mx} dx = \frac{e^{mx}}{m} + C$$

$$6. \int \sin x dx = -\cos x + C$$

$$7. \int \cos x dx = \sin x + C$$

$$8. \int \sec^2 x dx = \tan x + C$$

$$9. \int \sec x \tan x dx = \sec x + C$$

$$10. \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$11. \int \operatorname{cosec} x \cdot \cot x dx = -\operatorname{cosec} x + C$$

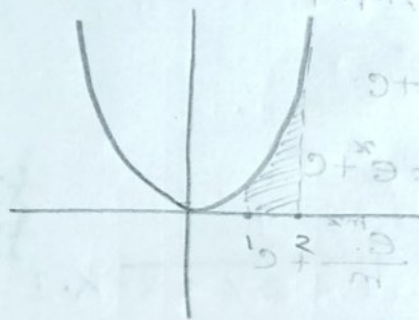
$$12. \int \frac{1}{b^x} dx = \frac{1}{\ln b} + C$$

Ex-1

Find the area of $f(x) = x^2$ over $[1, 2]$

Solⁿ:

$$\begin{aligned} & \int_1^2 x^2 dx \\ &= \left[\frac{x^3}{3} \right]_1^2 \\ &= \frac{8}{3} - \frac{1}{3} = \frac{7}{3} \end{aligned}$$



Not continuous
at $x=0$
So, impossible

Definite Integral:

If the integral of integration is defined, then this integral is called definite integral.

$$A = \int_a^b f(x) dx$$

$$= F(b) - F(a)$$

Indefinite Integral:

If the integral of integration is "Not" defined, then this integral is called indefinite integral.

$$A = \int f(x) dx$$

$$= F(x) + C ; \text{ where, } C \text{ is a constant.}$$

• Some Properties :

$$1. \int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$2. \int_a^a f(x) dx = 0$$

$$3. \int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$

$$4. \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$5. \text{ If } f(x) \geq 0 \quad \forall x \text{ in } [a, b]$$

$$\text{then, } \int_a^b f(x) dx \geq 0$$

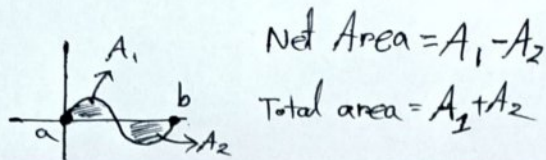
$$6. \text{ If } f(x) \geq g(x) \quad \forall x \text{ in } [a, b]$$

$$\text{then, } \int_a^b f(x) dx \geq \int_a^b g(x) dx$$



• $\forall$ = for all values of...

$$|x| = \begin{cases} x & ; x \geq 0 \\ -x & ; x < 0 \end{cases}$$



Total Area:

If f is continuous on $[a, b]$ then the total area is

$$A = \int_a^b |f(x)| dx$$

Ex: Find the total area of $f(x) = 1 - x^2$ over $[0, 2]$

Solⁿ:

$$A = \int_0^2 |f(x)| dx = \int_0^2 |1 - x^2| dx \quad \text{--- (i)}$$

Now,

$$|f(x)| = |1 - x^2| = \begin{cases} (1 - x^2); & 0 \leq x < 1 \\ -(1 - x^2); & 1 < x \leq 2 \end{cases}$$

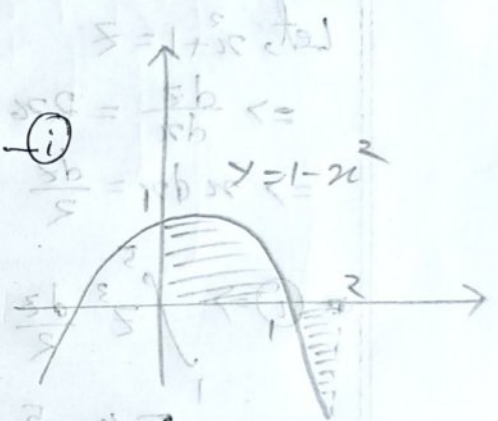
$$\text{(i)} \Rightarrow A = \int_0^1 f(x) dx + \int_1^2 f(x) dx$$

$$= \int_0^1 (1 - x^2) dx + \int_1^2 -(1 - x^2) dx$$

$$= \left[x - \frac{x^3}{3} \right]_0^1 + \left[-x + \frac{x^3}{3} \right]_1^2$$

$$= \frac{2}{3} + \frac{4}{3}$$

$$= 2$$



HW $\rightarrow$ Find the total area of $f(x) = 7 - x^2$ over $[-1, 2]$

Ans: 18

Anton: $5.2 \rightarrow (15 - 34)$
 $5.4 \rightarrow (27 - 48)$

Types of Integration:

Type-1: (Method of Substitution)

Ex-4): $\int_0^2 x(x^2+1)^3 dx$ — ①

Let, $x^2+1 = z$

$\Rightarrow \frac{dz}{dx} = 2x$

$\Rightarrow x dx = \frac{dz}{2}$

$\Rightarrow \textcircled{1} \Rightarrow \int_1^5 z^3 \frac{dz}{2}$

$= \frac{1}{2} \left[\frac{z^4}{4} \right]_1^5$

$= \frac{1}{2} \left[\frac{5^4}{4} - \frac{1}{4} \right]$

$= 78$

x	0	2
z	1	5

$f(x) = z$ if,

1. $[f(x)]^n$

2. $\frac{1}{\{f(x)\}^n}$

3. $\sqrt{f(x)}$

4. $\sqrt{\{f(x)\}^n}$

5. $\sqrt[n]{f(x)}$

6. $\sqrt[n]{\{f(x)\}^k}$

Ex-2:

$$\int \frac{dx}{(x-8)^5}$$

Let,

$$x-8 = z$$

$$\Rightarrow dx = dz$$

$$\therefore \int \frac{dx}{(x-8)^5} = \int \frac{dz}{z^5}$$

$$= \int z^{-5} \cdot dz$$

$$= \frac{z^{-4}}{-4} + C$$

$$= \frac{(x-8)^{-4}}{-4} + C$$

$$= -\frac{1}{4} \cdot \frac{1}{(x-8)^4} + C$$

HW

$$1. \int_0^{\pi/8} \sin^5 2x \cdot \cos 2x \, dx \quad \frac{1}{96}$$

$$2. \int_0^5 (2x-5)(x-3)^9 \, dx \quad \frac{5233}{110}$$

3. Anton: 2
Chapter: 5.9 $\rightarrow$ (5-18)
Chapter: 7.2 $\rightarrow$ (1-12)

$$4. \int \frac{dx}{(\frac{1}{3}x-8)^5} \quad \$$$

$$5. \int \frac{\sin \sqrt{x}}{\sqrt{x}} \, dx$$

$$6. \int \frac{dx}{1+e^{2x}}$$

$$7. \int x^3 \sqrt{x^4+5} \, dx$$

$$8. \int x^2 \sqrt{x-1} \, dx$$

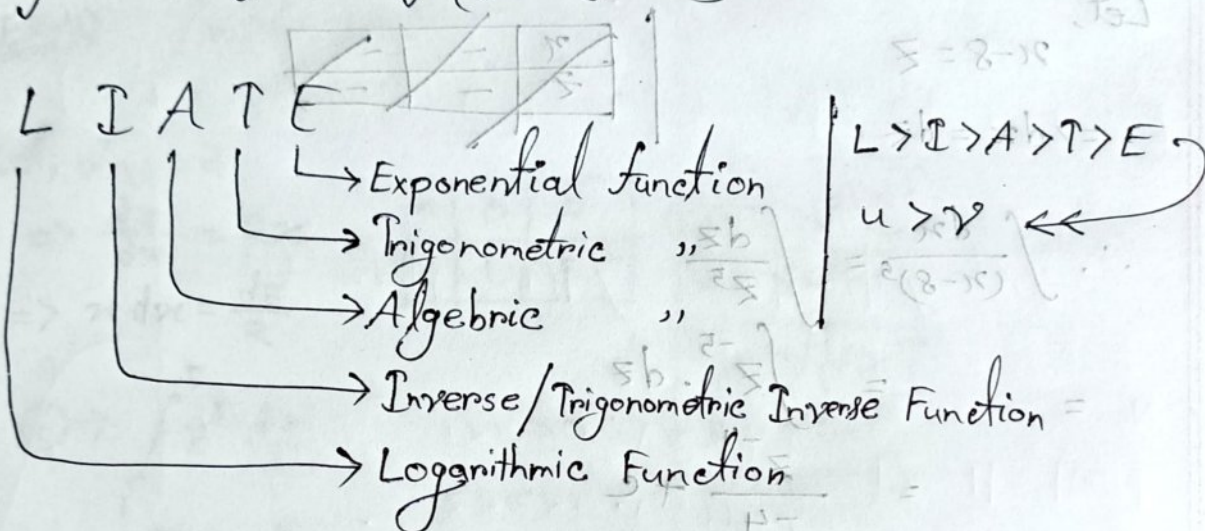
$$9. \int \frac{\sin(\ln x)}{x} \, dx$$

$$10. \int (x^2+1)^2 2x \, dx$$

Type-2: (Integration by Parts)

$$\int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx$$

L I A T E



HW

1. $\int x \tan^{-1} x dx$

2. $\int x \tan^2 x dx$

3. $\int e^{3x} \cos 2x dx$

4. $\int_0^{\pi} (x + x \cos x) dx$

5. Anton:

5.3 $\rightarrow (15-56)$

7.2 $\rightarrow (10-35)$

Ex: $\int e^x \sin 2x \cdot dx$

Solⁿ: $I = \int e^x \sin 2x \cdot dx$ ——— (i)

$$\Rightarrow I = \underbrace{e^x}_u \underbrace{\int \sin 2x \cdot dx}_{\int v} - \int \left\{ \frac{d}{dx} (\underbrace{e^x}_u) \underbrace{\int \sin 2x \cdot dx}_{\int v} \right\} dx$$

$$= e^x \left(\frac{-\cos 2x}{2} \right) - \int \left[e^x \cdot \frac{-\cos 2x}{2} \right] dx$$

$$= -\frac{1}{2} e^x \cos 2x + \frac{1}{2} \int \underbrace{e^x}_u \underbrace{\cos 2x}_{v} dx$$

$$I = -\frac{1}{2} e^x \cos 2x + \frac{1}{2} \left[e^x \int \cos 2x dx - \int \frac{d}{dx} (e^x) \int \cos 2x \right] dx$$

$$= -\frac{1}{2} e^x \cos 2x + \frac{1}{2} \left[e^x \cdot \frac{\sin 2x}{2} - \int e^x \cdot \frac{\sin 2x}{2} dx \right]$$

$$I = -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x - \frac{1}{4} \int e^x \cdot \sin 2x \cdot dx$$

$\underbrace{\hspace{10em}}_I$

$$\Rightarrow I = -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x - \frac{1}{4} I \quad [\text{From (i)}]$$

$$\Rightarrow I + \frac{1}{4} I = -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x$$

$$\Rightarrow \frac{5I}{4} = -\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x$$

$$\therefore I = \frac{4}{5} \cdot e^x \left[-\frac{1}{2} \cos 2x + \frac{1}{4} \sin 2x \right]$$

$$I = \frac{4}{5} \left[-\frac{1}{2} e^x \cos 2x + \frac{1}{4} e^x \sin 2x \right] \quad (\underline{\underline{\text{Ans:-}}})$$

Till

Quiz-1 : 24-06-2024 (Lec-4) $\Rightarrow$ 15 marks
(2/3 question)
(no opt.)

Type-3: Reduction Formula:

Obtain the Reduction formula for $\int \sin^n x \, dx$

Hence, evaluate, $\int \sin^6 x \, dx$

Solⁿ:

$$I_n = \int \sin^n x \cdot dx$$

$$= \int (\sin^{n-1} x) (\sin x) \, dx$$

$$= \sin^{n-1} x \int \sin x \, dx - \int \left\{ \frac{d}{dx} (\sin^{n-1} x) \int \sin x \, dx \right\} dx$$

$$= \sin^{n-1} x (-\cos x) - \int \{ (n-1) \cdot \sin^{n-2} x \cdot \cos x \cdot (-\cos x) \} dx$$

$$= -\sin^{n-1} x \cdot \cos x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx$$

$$= -\sin^{n-1} x \cdot \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx$$

$$= -\sin^{n-1} x \cdot \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

$$\Rightarrow I_n = -\sin^{n-1} x \cdot \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) I_n \quad [\text{From (i)}]$$

$$\Rightarrow I_n + (n-1) I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$$

$$\Rightarrow I_n (1+n-1) = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$$

$$\Rightarrow I_n \cdot n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$$

$$\therefore * \quad I_n = \frac{-\sin^{n-1} x \cos x + (n-1) I_{n-2}}{n} \quad *$$

for, $\int \sin^n x \, dx$

HW

$$\int \cos^n x \, dx \quad \checkmark$$

$$\int \tan^n x \, dx \quad \checkmark$$

$$\int \sec^n x \, dx \quad \checkmark$$

From formula,

$$I_6 = \frac{-\sin^{6-1} x \cos x + (6-1) I_{6-2}}{6}$$

$$= \frac{I_4}{6}$$

Now, for $I_6 = \int \sin^6 x \, dx$

$$I_6 = \frac{-\sin^{6-1} x \cos x + (6-1)I_{6-2}}{6}$$

$$= (-\sin^5 x \cos x + 5I_4) \cdot \frac{1}{6}$$

Again for I_4 , $\rightarrow$ same Formula.

$$\therefore I_6 = \frac{1}{6} \left[-\sin^5 x \cos x + 5 \left(\frac{-\sin^3 x \cos x + 3I_2}{4} \right) \right]$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{15}{24} I_2$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{15}{24} \left[\frac{-\sin x \cos x + I_0}{2} \right]$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x - \frac{15}{48} \sin x \cos x + \frac{15}{48} \int \sin^0 x \, dx$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x - \frac{15}{48} \sin x \cos x + \frac{15}{48} x + C$$

• HW solⁿ:

$$I_n = \frac{\cos^{n-1} x \cdot \sin x + (n-1)I_{n-2}}{n} ; \int \cos^n x \cdot dx \leftarrow (1)$$

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2} ; \int \tan^n x \cdot dx$$

$$I_n = \frac{\sec^{n-2} x \tan x + (n-2)I_{n-2}}{n-1} ; \int \sec^n x \, dx$$

Type-4: (Trigonometric Integration)

Integrating products of sine & cosine

$\int \sin^m x \cos^n x dx$, where, $m, n \in \mathbb{Z}$

• Case-1: If 'm' or 'n' is odd, then split the odd power

using $\left. \begin{aligned} \sin^2 x &= 1 - \cos^2 x \\ \cos^2 x &= 1 - \sin^2 x \end{aligned} \right\}$

Ex: $\int \sin^3 x \cos^4 x dx$

$\int \sin^2 x \cdot \sin x \cos^4 x dx$

$= \int (1 - \cos^2 x) \sin x \cos^4 x \cdot dx$ — (i)

Let, $\cos x = z$

$\Rightarrow -\sin x dx = dz$ — (ii)

(i) $\Rightarrow$ (i)

$-\int (1 - z^2) z^4 dz$

$= \int (z^6 - z^4) dz$

$= \frac{z^7}{7} - \frac{z^5}{5} + C$

$= \frac{\cos^7 x}{7} - \frac{\cos^5 x}{5} + C$ (Ans:-)

• Case-2: If 'm' and 'n' are odd, then split either ^{any} of them

using -

$$\sin^2 x = 1 - \cos^2 x$$

$$\cos^2 x = 1 - \sin^2 x$$

Ex: $\int \sin^3 x \cos^3 x \, dx$

$$= \int \sin^2 x \cdot \cos x \cdot \cos^2 x \cdot dx$$

$$= \int \sin^2 x \cdot \cos x (1 - \sin^2 x) \, dx$$

Let,

$$\sin x = z$$

$$\Rightarrow \cos x \, dx = dz \quad \text{--- (ii)}$$

$$(ii) \Rightarrow (i)$$

$$\int z^3 (1 - z^2) \, dz$$

$$= \int (z^3 - z^5) \, dz$$

$$= \frac{z^4}{4} - \frac{z^6}{6} + C$$

$$= \frac{\sin^4 x}{4} - \frac{\sin^6 x}{6} + C$$

$$\begin{aligned} 2\sin^2 x &= 1 - \cos 2x \\ 2\cos^2 x &= 1 + \cos 2x \end{aligned}$$

Case-3: If 'm' & 'n' are even, then use $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and, finally use the Reduction formula. $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

Ex: $\int \sin^6 x \cos^4 x \, dx$

$$= \int (\sin^2 x)^3 (\cos^2 x)^2 \, dx$$

$$= \int \left\{ \frac{1}{2}(1 - \cos 2x) \right\}^3 \left\{ \frac{1}{2}(1 + \cos 2x) \right\}^2 \, dx$$

$$= \int \frac{1}{64} (1 - \cos 2x)^3 (1 + \cos 2x)^2 \, dx$$

$$= \frac{1}{64} \int (1 - \cos^2 2x)^3 \, dx$$

$$= \frac{1}{64} \int (\sin^2 2x)^3 \, dx$$

$$= \frac{1}{64} \int \sin^6(2x) \, dx \quad \text{--- (i)}$$

Let, $2x = z$

$$\Rightarrow 2 \, dx = dz$$

$$\Rightarrow dx = \frac{dz}{2} \quad \text{--- (ii)}$$

$$(i) \Rightarrow (i),$$

$$\frac{1}{64} \int \sin^6 z \cdot \frac{dz}{2}$$

$$= \frac{1}{128} \int \sin^6 z \cdot dz$$

⋮

$$= \frac{1}{128} \int \sin^6 z \, dz$$

$$= \frac{1}{128} [I_6] \quad [I_n \rightarrow z \text{ দিয়ে কমানো হবে}]$$

$$= \dots \text{ (Reduction Formula)}$$

Derive করে

Ans. এর ক্যাণ্ডে ২৬

(3-4) Page হয়ে গছে

#2 $\int \tan^n x \sec^m x dx$

• Case-1: If the power of $\sec x$ is even, then split.

$$\sec^2 x = 1 + \tan^2 x$$

Ex: $\int \tan^3 x \sec^5 x dx$

$$= \int \tan^3 x (\sec^2 x) (\sec^3 x) dx$$

$$= \int \tan^3 x \sec^3 x (1 + \tan^2 x) dx \quad \text{--- (i) Let, } \tan x = z$$

$$\Rightarrow \sec^2 x \cdot dx = dz$$

$$= \int z^3 (1 + z^2) \cdot dz$$

$$= \frac{z^4}{4} + \frac{z^6}{6} + C$$

$$= \frac{\tan^4 x}{4} + \frac{\tan^6 x}{6} + C$$

• Case-2: If the power of $\sec x$ and $\tan x$ are odd, then split $\boxed{\sec x \tan x}$

Ex: $\int \tan^5 x \sec^3 x \cdot dx$

$$= \int \tan^4 x \sec^2 x \cdot \sec x \tan x \, dx$$

$$= \int (\tan^2 x)^2 \sec^2 x \cdot \sec x \tan x \, dx$$

$$= \int (\sec^2 x - 1)^2 \sec^2 x \cdot \sec x \tan x \, dx \quad \text{--- (1)} \quad \left| \begin{array}{l} \text{Let, } \sec x = z \\ \Rightarrow \sec x \tan x \cdot dx = dz \end{array} \right.$$

$$= \int (z^2 - 1)^2 \cdot z^2 \, dz$$

$$= \int (z^4 - 2z^2 + 1) z^2 \, dz$$

$$= \int (z^6 - 2z^4 + z^2) \, dz$$

$$= \frac{z^7}{7} - \frac{2z^5}{5} + \frac{z^3}{3} + C$$

$$= \frac{\sec^7 x}{7} - \frac{2\sec^5 x}{5} + \frac{\sec^3 x}{3} + C \quad \text{(Ans:-)}$$

• Case-3: If the power of $\tan x$ is even (and $\sec x$ is odd) then, express in $\sec x$ and then apply the Reduction formula

Ex-1

$$\int \tan^4 x \sec x \, dx$$

$$= \int (\tan^2 x)^2 \cdot \sec x \, dx \quad \text{--- (1)}$$

$$= \int (\sec^2 x - 1)^2 \cdot \sec x \, dx \quad \text{--- (2)}$$

$$= \int (\sec^4 x - 2\sec^2 x - 1) \sec x \, dx \quad \text{--- (3)}$$

$$= \int (\sec^5 x - 2\sec^3 x - \sec x) \, dx$$

$$= \int \sec^5 x \, dx - 2 \int \sec^3 x \, dx - \int \sec x \, dx$$

$$= \left(\text{Reduction formula for } \int \sec^5 x \, dx \right) - 2 \left(\text{Reduction formula for } \int \sec^3 x \, dx \right) - \int \sec x \, dx$$

$$= \{ I_5 + 2 I_3 + I \}$$

→ Derive + then, formula if needed

Ans

$$= \int \sec^5 x \, dx - 2 \int \sec^3 x \, dx - \int \sec x \, dx$$

$$= \int \sec^3 x \, dx + \frac{\sec x \tan x}{2} - \frac{1}{2} \int \sec x \, dx - 2 \left(\int \sec x \, dx + \frac{\sec x \tan x}{2} - \frac{1}{2} \int \sec x \, dx \right) - \int \sec x \, dx$$

$$= \int \sec^3 x \, dx + \frac{\sec x \tan x}{2} - \frac{1}{2} \int \sec x \, dx - \int \sec x \, dx - \frac{\sec x \tan x}{2} + \frac{1}{2} \int \sec x \, dx - \int \sec x \, dx$$

$$= \int \sec^3 x \, dx - \int \sec x \, dx - \frac{\sec x \tan x}{2} + \frac{1}{2} \int \sec x \, dx$$

$$= \int \sec^3 x \, dx - \frac{1}{2} \int \sec x \, dx - \frac{\sec x \tan x}{2}$$

$$= \int \sec^3 x \, dx - \frac{1}{2} \int \sec x \, dx - \frac{\sec x \tan x}{2}$$

Type-5: (Integral of the form)

① $\int \sin mx \cdot \cos nx \, dx$, $\int \sin mx \cdot \sin nx \, dx$, $\int \cos mx \cdot \cos nx \, dx$
can be found by using -

$$① \sin a \cdot \cos b = \frac{1}{2} [\sin(a+b) + \sin(a-b)]$$

$$② \sin a \cdot \sin b = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

$$③ \cos a \cdot \cos b = \frac{1}{2} [\cos(a+b) + \cos(a-b)]$$

Ex-1 $\int \sin 4x \sin 9x \, dx$

$$= \int \frac{1}{2} [\cos(4x-9x) - \cos(4x+9x)] \, dx$$

$$= \int \frac{1}{2} [\cos(-5x) - \cos(13x)] \, dx$$

$$= \frac{1}{2} \int (\cos 5x - \cos 13x) \, dx$$

$$= \frac{1}{2} \left(\frac{\sin 5x}{5} - \frac{\sin 13x}{13} \right) + C$$

Ex-2:

$$\int \sin 7x \cos 3x \, dx$$

$$= \int \frac{1}{2} [\sin(7x+3x) + \sin(7x-3x)] \, dx$$

$$= \int \frac{1}{2} (\sin 10x + \sin 4x) \, dx$$

$$= \frac{1}{2} \left[\left(-\frac{\cos 10x}{10} \right) + \left(-\frac{\cos 4x}{4} \right) \right] + C$$

$$= -\frac{1}{2} \left[\frac{\cos 10x}{10} + \frac{\cos 4x}{4} \right] + C$$

Type-6: (Trigonometric Substitution)

We know that,

$$\textcircled{i} \sec^2 \theta = 1 + \tan^2 \theta$$

$$\textcircled{ii} \tan^2 \theta = \sec^2 \theta - 1$$

$$\textcircled{iii} \sin^2 \theta = 1 - \cos^2 \theta$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

Format: $\int \sqrt{a^2 + x^2} dx$

Let, $x = a \tan \theta$

$$\begin{aligned} & \therefore \int \sqrt{a^2 + (a \tan \theta)^2} d\theta \\ &= \int \sqrt{a^2 (1 + \tan^2 \theta)} d\theta \\ &= \int a \cdot \sec \theta d\theta \\ &= \int a \cdot \sec \theta d\theta \end{aligned}$$

Format: $\int \sqrt{x^2 - a^2} dx$

Let, $x = a \sec \theta$

$$\begin{aligned} & \therefore \int \sqrt{a^2 \sec^2 \theta - a^2} d\theta \\ &= \int a \sqrt{\sec^2 \theta - 1} d\theta \\ &= \int a \tan \theta d\theta \end{aligned}$$

Format: $\int \sqrt{a^2 - x^2} dx$

Let, $x = a \cos \theta$ (or) $x = a \sin \theta$ (any one)

$$\begin{aligned} & \therefore \int \sqrt{a^2 - a^2 \cos^2 \theta} d\theta \quad \left| \int \sqrt{a^2 - a^2 \sin^2 \theta} d\theta \right. \\ &= \int a \sin \theta d\theta \quad \left| \int a \cos \theta d\theta \right. \end{aligned}$$

Ex-1: $\int \frac{dx}{x^2 \sqrt{9-x^2}}$ — (i)

Sol:

Let,

$$x = 3 \sin \theta$$

$$dx = 3 \cos \theta d\theta$$

$$\textcircled{i} \Rightarrow \int \frac{3 \cos \theta d\theta}{9 \sin^2 \theta \sqrt{9 - 9 \sin^2 \theta}}$$

$$= \int \frac{1}{3} \cdot \frac{\cos \theta d\theta}{\sin^2 \theta \sqrt{9(1 - \sin^2 \theta)}}$$

$$= \int \frac{1}{9} \cdot \frac{\cos \theta d\theta}{\sin^2 \theta \cos \theta}$$

$$= \frac{1}{9} \cdot \int \operatorname{cosec}^2 \theta d\theta$$

$$= -\frac{1}{9} \cot \theta + C \quad \text{--- (ii)}$$

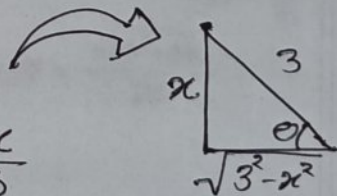
$$x = 3 \sin \theta$$

$$\Rightarrow \sin \theta = \frac{x}{3}$$

$$\therefore \cot \theta = \frac{\sqrt{9-x^2}}{x}$$

$$\textcircled{ii} \Rightarrow -\frac{1}{9} \left(\frac{\sqrt{9-x^2}}{x} \right) + C$$

(Ans:-)



Ex-2: $\int \frac{dx}{x^2 \sqrt{x^2+25}}$ — (i)

Solⁿ: Let, $x = 5 \tan \theta$
 $\Rightarrow dx = 5 \sec^2 \theta \cdot d\theta$

(i) $\Rightarrow \int \frac{5 \sec^2 \theta \cdot d\theta}{25 \tan^2 \theta \sqrt{25 \tan^2 \theta + 25}}$

$= \int \frac{5 \sec^2 \theta \cdot d\theta}{(5 \tan \theta)^2 \sqrt{5^2 \cdot \sec^2 \theta}}$

$= \int \frac{5 \sec^2 \theta \cdot d\theta}{25 \tan^2 \theta \cdot 5 \sec \theta}$

$= \frac{1}{25} \int \frac{\cos^3 \theta}{\cos \theta \cdot \sin^2 \theta} \cdot d\theta$

$= \frac{1}{25} \int \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} d\theta$

$= \frac{1}{25} \int \cot \theta \cdot \operatorname{cosec} \theta \cdot d\theta$

$= \frac{1}{25} (-\operatorname{cosec} \theta) + C$ — (ii)

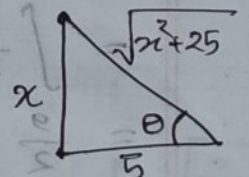
$= \frac{1}{25} \left(-\frac{\sqrt{x^2+25}}{x} \right) + C$

$= -\frac{\sqrt{x^2+25}}{25x} + C$

$x = 5 \tan \theta$

$\therefore \tan \theta = \frac{x}{5}$

$\therefore \operatorname{cosec} \theta = \frac{\sqrt{x^2+25}}{x}$



Ex-3: $\int \frac{x^2}{\sqrt{9-x^2}} dx$ — (i)

Sol: Let, $x = 3 \sin \theta$
 $dx = 3 \cos \theta \cdot d\theta$

(i) $\Rightarrow \int \frac{9 \sin^2 \theta}{\sqrt{9-9 \sin^2 \theta}} \cdot 3 \cos \theta \cdot d\theta$

$= \int \frac{3 \cdot 9 \cdot \sin^2 \theta \cos \theta}{3 \cos \theta} d\theta$

$= \int 9 \sin^2 \theta d\theta$

$= \int \frac{9}{2} \cdot 2 \sin^2 \theta \cdot d\theta$

$= \frac{9}{2} \int (1 - \cos 2\theta) \cdot d\theta$

$= \frac{9}{2} \cdot \left[\theta - \frac{\sin 2\theta}{2} \right] + C$ — (ii)

$= \frac{9}{2} \left[\sin^{-1} \frac{x}{3} - \frac{1}{2} \sin \left(\sin^{-1} \frac{x}{3} \right) \right] + C$

$\frac{x^2}{\sqrt{9-x^2}} \int \frac{x^2}{\sqrt{9-x^2}}$

Let $x = 3 \sin \theta$

ob. $\frac{dx}{d\theta} = 3 \cos \theta$

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② (Square roots of Quadratics):

Ex-1 $\int \sqrt{4x^2 - 16x + 52} dx$

$$= \int \sqrt{4(x^2 - 4x + 13)} dx$$

$$= 2 \int \sqrt{x^2 - 4x + 13} dx$$

$$= 2 \int \sqrt{(x)^2 - 2 \cdot x \cdot 2 + 2^2 + 9} dx$$

$$= \int 2 \sqrt{(x-2)^2 + 3^2} dx \quad \text{--- (ii) Let,}$$

$$= \int 2 \sqrt{z^2 + 3^2} dz \quad \text{--- (iii) } \begin{matrix} x-2=z \\ dz=dx \end{matrix}$$

Let,

$$z = 3 \tan \theta$$

$$\Rightarrow dz = 3 \sec^2 \theta \cdot d\theta$$

$$\text{(iv)} \Rightarrow \int 2 \cdot \sqrt{9 \tan^2 \theta + 9} \cdot 3 \sec^2 \theta \cdot d\theta$$

$$= \int 2 \cdot 3 \cdot \sec \theta \cdot 3 \sec^2 \theta \cdot d\theta$$

$$= \int 18 \sec^3 \theta \cdot d\theta$$

$$= 18 \int \sec^3 \theta \cdot d\theta$$

$\int \sec^3 \theta \cdot d\theta = I_n$
(Use Reduction Formula) $\rightarrow$ (Directly use formula, don't need to derive)

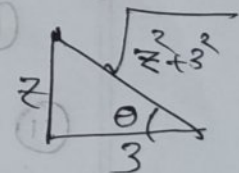
$$\Rightarrow 18 \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right] + C$$

--- (iv)

$$\tan \theta = \frac{z}{3}$$

$$\Rightarrow \sec \theta = \frac{\sqrt{z^2 + 3^2}}{3}$$

$$\text{(iv)} \Rightarrow 18 \left[\frac{1}{2} \cdot \frac{\sqrt{(x-2)^2 + 9}}{3} \cdot \frac{x-2}{3} + \frac{1}{2} \ln \left| \frac{\sqrt{9 + (x-2)^2}}{3} + \frac{x-2}{3} \right| \right] + C$$



Ex-2:

$$\int \sqrt{x^2 - 5x + 4} dx$$

$$= \int \sqrt{2(x^2 - \frac{5}{2}x + 2)} dx$$

$$= \sqrt{2} \int \sqrt{(x^2 - 2 \cdot x \cdot \frac{5}{4} + (\frac{5}{4})^2) + 2 - (\frac{5}{4})^2} dx$$

$$= \sqrt{2} \int \sqrt{(x - \frac{5}{4})^2 + \dots}$$

(Same as before)
previous

HW

① $\int x^3 \sqrt{5-x^2} dx$; $x = \sqrt{5} \sin \theta$

② $\int \frac{3x^3 dx}{\sqrt{x^2-25}}$; $x = 5 \sec \theta$

③ $\int \frac{dx}{x^2 \sqrt{4-x^2}}$; $x = 2 \sin \theta$

④ $\int \frac{\sqrt{4x^2-25}}{x} dx$; $x = 5 \sec \theta$

(Use Reduction Formulae)
need to derive formulae

⑤ $\int \frac{1}{x^2 \sqrt{4-x^2}} dx$

Improper Integral:

Consider the definite integral $\int_a^b f(x) \cdot dx$

If the integrated $f(x)$ has point of discontinuity on $[a, b]$ or any or both a & b , then the integral is called - Improper integral.

Ex(1): $\int_1^{\infty} \frac{dx}{x^2}$, $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$, $\int_{-\infty}^1 e^x dx$ $\rightarrow$ these are improper integral with infinite interval

Ex(2): $\int_{-1}^1 \frac{dx}{x^2}$, $\int_1^2 \frac{dx}{x-1}$ $\rightarrow$ these are improper integral with discontinuity

Ex(3): $\int_0^{\infty} \frac{dx}{\sqrt{x}}$ $\rightarrow$ This is improper integral with infinite discontinuity & infinite interval

How to evaluate improper integral?

Solⁿ:

$$\int_a^b f(x) dx$$

D.N.E $\rightarrow$ Does Not Exist

$$(1) \int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

$$(2) \int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

If the limit exists, then the improper integral is said to converge. If the limit D.N.E. then the integral is said to Diverge.

$\Downarrow$
if limit exist $\equiv$ finite value
then, Converge
& if limit doesn't exist $\equiv$ infinite value
then, Diverge

(Ex-1) $\int_1^\infty \frac{dx}{x}$

$$= \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x}$$

$$= \lim_{b \rightarrow \infty} [\ln x]_1^b$$

$$= \lim_{b \rightarrow \infty} [\ln b - \ln 1]$$

$$= \infty$$

Thus, the integral is diverge. (Ans:)

Ex-2: $\int_2^{\infty} \frac{dx}{(x-1)^2}$

$$= \lim_{b \rightarrow \infty} \int_2^b \frac{dx}{(x-1)^2}$$

$$= \lim_{b \rightarrow \infty} \left[\frac{-1}{x-1} \right]_2^b$$

$$= \lim_{b \rightarrow \infty} \left(\frac{-1}{b-1} + \frac{1}{2-1} \right)$$

$$= 0 + 1$$

$$= 1$$

Thus, the integral is converge.

Ex-3: $\int_0^{\infty} (1-x) e^x dx$

$$= \lim_{b \rightarrow \infty} \int_0^b (1-x) e^x dx$$

Now

$$\int_0^b (1-x) e^x dx = (1-x) \int_0^b e^x dx - \int_0^b \left\{ \frac{d}{dx} (1-x) \int_0^b e^x dx \right\} dx$$

$$= (1-x) [e^x]_0^b - \int_0^b (-1)(-1)(e^x) dx$$

$$= -(1-x) [e^x]_0^b + \int_0^b [e^x]_0^b$$

$$= -(1-x)(e^b - 1) + (e^b - 1)$$

$$=$$

Ex-3 : $\int_0^{\infty} (1-x) e^{-x} dx$

$$= \lim_{b \rightarrow \infty} \int_0^b (1-x) e^{-x} dx$$

Now,

$$\int_0^b (1-x) e^{-x} dx = (1-x) \int_0^b e^{-x} dx - \int_0^b \left\{ \frac{d}{dx} (1-x) \int_0^b e^{-x} dx \right\} dx$$

$$= \left[-(1-x) e^{-x} \right]_0^b - \int_0^b \{ (-1)(e^{-x}) \} dx$$

$$= \left[-(1-b) e^{-b} + 1 \right] + \left[e^{-x} \right]_0^b$$

$$= -e^{-b} + b e^{-b} + 1 + \left[e^{-b} - 1 \right]$$

$$= -e^{-b} + b e^{-b} + 1 + e^{-b} - 1$$

$$= b \cdot e^{-b}$$

$$\therefore \lim_{b \rightarrow \infty} \int_0^b (1-x) e^{-x} dx = \lim_{b \rightarrow \infty} b \cdot e^{-b}$$

$$= \lim_{b \rightarrow \infty} \frac{b}{e^b} \rightarrow \left[\frac{\infty}{\infty} \right]$$

$$= \lim_{b \rightarrow \infty} \frac{1}{e^b} \quad \text{[Using L'Hôpital's Rule]}$$

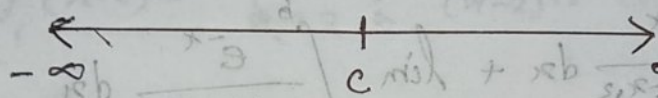
$$= 0$$

Thus, the integral is converge.

Tip: $\int f(x)$, use integration by parts with no limit, then use limit after calculating everything.

□ The improper integral of the form

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$$



where, c is any real number. If both term is converge, then, the integral is said to converge, otherwise diverge.

$$\int_{-\infty}^{\infty} f(x) dx = \lim_{a \rightarrow -\infty} \int_a^c f(x) dx + \lim_{b \rightarrow \infty} \int_c^b f(x) dx$$

if both $\rightarrow$ finite + finite

then, converge;

otherwise, diverge.

Ex(4) $\int_{-\infty}^{\infty} \frac{e^{-x}}{1+e^{-2x}} dx$

$$= \int_{-\infty}^0 \frac{e^{-x}}{1+e^{-2x}} dx + \int_0^{\infty} \frac{e^{-x}}{1+e^{-2x}} dx$$

$$= \lim_{a \rightarrow -\infty} \int_a^0 \frac{e^{-x}}{1+(e^{-x})^2} dx + \lim_{b \rightarrow \infty} \int_0^b \frac{e^{-x}}{1+(e^{-x})^2} dx$$

~~Let~~ Now, the integral is said to converge if

$$\int \frac{e^{-x}}{1+(e^{-x})^2} dx$$

$\Rightarrow$ As both ~~the~~ function is same, we will calculate once with no limit, and then, use limit for each term.

Let,

$$e^{-x} = u$$

$$\Rightarrow -e^{-x} dx = du$$

$$\therefore \int \frac{-du}{1+u^2}$$

$$= -\tan^{-1} u$$

$$= -\tan^{-1}(e^{-x})$$

$$\left| \begin{array}{l} e^{+\infty} = \infty \\ e^{-\infty} = 0 \end{array} \right.$$

Now, $\lim_{a \rightarrow -\infty} [-\tan^{-1}(e^{-x})]_a^0 + \lim_{b \rightarrow +\infty} [-\tan^{-1}(e^{-x})]_0^b$

$$= \lim_{a \rightarrow -\infty} [-\tan^{-1}(1) + \tan^{-1}(e^a)] + \lim_{b \rightarrow +\infty} [-\tan^{-1}(e^{-b}) + \tan^{-1}(1)]$$

$$= -\frac{\pi}{4} + \frac{\pi}{2} + (-0) + \frac{\pi}{4} = \frac{\pi}{2}$$

Ex 5. $\int_1^4 \frac{dx}{(x-2)^{2/3}}$

$$= \int_1^2 \frac{dx}{(x-2)^{2/3}} + \int_2^4 \frac{dx}{(x-2)^{2/3}}$$

$$= \lim_{b \rightarrow 2} \int_1^b \frac{dx}{(x-2)^{2/3}} + \lim_{a \rightarrow 2} \int_a^4 \frac{dx}{(x-2)^{2/3}}$$

Now,

$$\int \frac{dx}{(x-2)^{2/3}} = 3(x-2)^{1/3} + C$$

$$= 3(x-2)^{1/3} + C$$

Now,

$$\lim_{b \rightarrow 2} \left[3(x-2)^{1/3} \right]_1^b + \lim_{a \rightarrow 2} \left[3(x-2)^{1/3} \right]_a^4$$

$$= [0 - (-3)] + [3 \cdot 2^{1/3} - 0]$$

$$= 3 + 3 \cdot 2^{1/3}$$

$$= 6.78$$

$$\frac{1}{\sqrt{2}} \sec \theta$$

HW

① $\int_1^2 \frac{dx}{1-x}$; (Diverge)

③ $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$; π (Converge)

② $\int_{-2}^2 \frac{dx}{x^2}$; (Diverge)

④ $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$; $\frac{\pi}{2}$ (Converge)

Lec-6

Quiz-2

10/7/2024

Topic: Lec-7 (Till)

1.7.2024

Monday

• Integrating Rational Functions by Partial Functions:

• Type-1: $\frac{1}{(x-a)(x-b)} \equiv \frac{A}{x-a} + \frac{B}{x-b}$

• Type-2: $\frac{1}{(x-a)(x-b)^2} \equiv \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}$

• Type-3: $\frac{1}{(x-a)(ax^2+bx+c)} \equiv \frac{A}{x-a} + \frac{Bx+C}{ax^2+bx+c}$

• Type-4: $\frac{1}{(x-a)(ax^2+bx+c)^2} \equiv \frac{A}{x-a} + \frac{Bx+C}{ax^2+bx+c} + \frac{Dx+E}{(ax^2+bx+c)^2}$

$\int \left[\frac{A}{x-a} + \frac{Bx+C}{ax^2+bx+c} \right] dx = \int \frac{A}{x-a} dx + \int \frac{Bx+C}{ax^2+bx+c} dx$

$\left[0 - \frac{1}{2} \cdot \frac{1}{2} \right] + \left[\frac{1}{2} - 0 \right] =$

$\frac{1}{2} + \frac{1}{2} =$

$1 \cdot 1 =$

HM

① $\int \frac{1}{x^2-1} dx$

② $\int \frac{1}{x^2+1} dx$

Ex-1: $\int \frac{5x-10}{x^2-3x-4} dx$

$$= \int \frac{5x-10}{(x-4)(x+1)} dx$$

Now,

$$\frac{5x-10}{(x-4)(x+1)} = \frac{A}{x-4} + \frac{B}{x+1} \quad \text{--- (i)}$$

$$\Rightarrow 5x-10 = A(x+1) + B(x-4) \quad \text{--- (ii)}$$

if $x=4$, (ii) $\Rightarrow A=2$

if $x=-1$, (ii) $\Rightarrow B=3$

Substituting A, B in (i),

$$\frac{5x-10}{(x-4)(x+1)} = \frac{2}{x-4} + \frac{3}{x+1}$$

Now,

$$\int \frac{5x-10}{(x-4)(x+1)} dx = \int \left[\frac{2}{x-4} + \frac{3}{x+1} \right] dx$$

$$= 2 \ln(x-4) + 3 \ln(x+1) + C$$

(Ans:)

Ex-2: $\int \frac{dx}{x^3 - x^2}$

$$= \int \frac{dx}{x^2(x-1)}$$

Now,

$$\frac{1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} \quad \text{--- (i)}$$

$$\Rightarrow 1 = A x(x-1) + B(x-1) + C x^2 \quad \text{--- (ii)}$$

if, $x=0$; (ii) $\Rightarrow B = -1$

$x=1$; (ii) $\Rightarrow C = 1$

Equating the co-efficient of x^2

$$C + A = 0$$

$$\Rightarrow A = -C$$

$$\Rightarrow A = -1$$

Now, Plugging A, B, C into (i),

$$\frac{1}{x^2(x-1)} = \frac{-1}{x} + \frac{-1}{x^2} + \frac{1}{x-1}$$

$$\therefore \int \frac{dx}{x^2(x-1)} = \int \left[\frac{-1}{x} + \frac{-1}{x^2} + \frac{1}{x-1} \right]$$

$$= -\ln x - \left(\frac{-1}{x} \right) + \ln(x-1) + C$$

$$= \ln(x-1) + \frac{1}{x} - \ln x + C$$

Ex-3: $\int \frac{x^2+x-2}{3x^3-x^2+9x-1} dx$

$$= \int \frac{x^2+x-2}{x^3(3x-1)+(3x-1)} dx$$

$$= \int \frac{x^2+x-2}{(3x-1)(x^3+1)} dx$$

Now, $\frac{x^2+x-2}{(3x-1)(x^3+1)} \equiv \frac{A}{(3x-1)} + \frac{Bx+C}{(x^3+1)} \quad \text{--- (i)}$

$$\Rightarrow x^2+x-2 \equiv A(x^3+1) + (Bx+C)(3x-1)$$

$$\Rightarrow x^2+x-2 \equiv Ax^3 + A + 3Bx^2 - Bx + 3Cx - C$$

$$\Rightarrow x^2+x-2 \equiv x^3(A+3B) + x(3C-B) + (A-C) \quad \text{--- (ii)}$$

Equating the co-efficient of x^3, x & constant term,

$$A+3B=1 \quad ; \quad 3C-B=1 \quad ; \quad A-C=-2$$

$$\text{--- (iii)} \Rightarrow 3A+6-B=1$$

$$\Rightarrow 3A-B=-5$$

$$\text{--- (iv)}$$

Solving (iii) (iv),

$$A = -\frac{7}{5}$$

$$B = \frac{4}{5}$$

$$\therefore C = -\frac{7}{5} + 2 = \frac{3}{5}$$

Plugging A, B, C in (i),

$$\frac{x^2+x-2}{(3x-1)(x^3+1)} \equiv \frac{-\frac{7}{5}}{(3x-1)} + \frac{\frac{4}{5}x + \frac{3}{5}}{(x^3+1)}$$

$$A+3B+C=1$$

Solve directly
in Calc.
(3 unknown variable)

Now

$$\int \left[-\frac{7}{5} \cdot \frac{1}{3x-1} + \frac{\frac{4}{5}x + \frac{3}{5}}{x^2+1} \right] dx$$

$$= \int \left[-\frac{7}{5} \cdot \frac{1}{3x-1} + \frac{4}{5} \cdot \frac{x}{x^2+1} + \frac{3}{5} \cdot \frac{1}{x^2+1} \right] dx$$

$$= -\frac{7}{5} \cdot \frac{\ln(3x-1)}{3} + \int \left(\frac{2}{5} \cdot \frac{2x}{x^2+1} + \frac{3}{5} \cdot \frac{1}{x^2+1} \right) dx$$

$$= -\frac{7 \ln(3x-1)}{15} + \frac{2}{5} \cdot \ln(x^2+1) + \frac{3}{5} \cdot \tan^{-1}x + C \quad (\text{Ans.})$$

Formula

$$\int \frac{f'(x)}{f(x)} dx = \ln f(x)$$

$$= \ln f(x)$$

(i)

$$(i) \rightarrow (C-A) + x(3C-B) + (A-C) = 5x^2 + x - 5$$

Equating the co-efficient of x^2 , x & constant term

$$A+3B=1 \quad ; \quad 3C-B=1 \quad ; \quad A-C=-5$$

$$\Rightarrow 3A+C=1 \quad (ii)$$

$$\Rightarrow 3A-B=-2 \quad (iii)$$

$$(iv)$$

Solving (iii) & (iv)

$$A = -\frac{x}{2}$$

$$B = \frac{1}{2}$$

$$\therefore C = -\frac{x}{2} + 5 = \frac{5}{2}$$

Putting A, B, C in (i)

$$\frac{5x^2+x-5}{(x^2+1)} = \frac{5x^2+x-5}{(x^2+1)} + \frac{\frac{x}{2}}{(x^2+1)} + \frac{\frac{1}{2}x + \frac{3}{2}}{(x^2+1)}$$

Solve directly in Calc. (8 marks available)

$\frac{x^a}{x^b}$ where, $a > b$; like $\frac{5}{3}$ $\rightarrow$ to convert to proper:

$$\frac{5}{3} = 1 + \frac{2}{3}$$

Improper Rational Function:

Ex(1): $\int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} dx$

Now,

$$3x^4 + 3x^3 - 5x^2 + x - 1$$

$$= 3x^2(x^2 + x - 2)$$

$$= 3x^4 + 3x^3 - 6x^2 + 3x^2 + x - 2 + 1$$

$$= 3x^2(x^2 + x - 2) + 1(x^2 + x - 2) + 1$$

$$= (3x^2 + 1)(x^2 + x - 2) + 1$$

$$= (x^2 + x - 2) \left[3x^2 + 1 + \frac{1}{x^2 + x - 2} \right]$$

$$\therefore \int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} = \frac{(x^2 + x - 2) \left[3x^2 + 1 + \frac{1}{x^2 + x - 2} \right]}{(x^2 + x - 2)}$$

$$= 3x^2 + 1 + \frac{1}{x^2 + x - 2}$$

$$\therefore \int \left(3x^2 + 1 + \frac{1}{x^2 + x - 2} \right) dx$$

$$= 3 \cdot \frac{x^3}{3} + x + \int \frac{1}{x^2 + x - 2} dx$$

$$= x^3 + x + \int \frac{1}{(x+2)(x-1)} dx$$

to convert to proper: $\frac{2}{3}$ like $\frac{2}{3}$ where, $\frac{2}{3}$ $\frac{2}{3}$

Improper Rational Function:

Now $\int \frac{1}{(x+2)(x-1)} dx$

Ex (1): $\int \frac{3x^2 + 3x - 2}{x^3 + x^2 - 2x} dx$

$\therefore \frac{1}{(x+2)(x-1)} = \frac{A}{(x+2)} + \frac{B}{(x-1)}$

At $x = -2 \Rightarrow 0 = A(x-1) + B(x+2)$

$0 \cdot x^2 + 0 \cdot x + 1 = Ax - A + Bx + 2B = x(A+B) + (2B-A)$

Equating the co-efficient of x and constant term:

$$\begin{array}{l|l} A+B=0 & 2B-A=1 \\ \Rightarrow A=-B & \Rightarrow 2B+B=1 \\ & \Rightarrow 3B=1 \\ & \Rightarrow B=\frac{1}{3} \end{array}$$

$(x-1)A = -\frac{1}{3}$

$A, B \rightarrow \textcircled{i}$

$\frac{1}{(x+2)(x-1)} = \frac{-\frac{1}{3}}{(x+2)} + \frac{\frac{1}{3}}{(x-1)} = \frac{-1}{3(x+2)} + \frac{1}{3(x-1)}$

$\int \frac{1}{x^3 + x^2 - 2x} dx = \int \frac{1}{x(x+2)(x-1)} dx$

$\int \frac{1}{(1-x)(x+2)} dx$

HW

$$1. \int \frac{dx}{x^2+x-2}$$

$$2. \int \frac{2x+4}{x^3-2x^2} dx$$

$$3. \int \frac{3x^4+4x^3+16x^2+20x+9}{(x+2)(x^2+3)^2} dx$$

$$4. \int \frac{5x^3-3x^2+7x-3}{(x^2+1)^2} dx$$

Anton: 7.5 $\rightarrow$ (1-34)

The Beta function is defined by definite integral

of the form

$$B(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$m, n > 0$$

Properties:

$$1. B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

$$2. \int_0^{\frac{\pi}{2}} \sin^n \theta \cos^n \theta d\theta = \frac{1}{2} B\left(\frac{n+1}{2}, \frac{n+1}{2}\right) = \frac{1}{2} \frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{\Gamma(n+1)}$$

• The Gamma and Beta Function:

The gamma function is defined by infinite integral of the form.

$$\Gamma n = \int_0^{\infty} e^{-x} \cdot x^{n-1} dx \longrightarrow \int_0^{\infty} e^{-x} x^5 dx$$

$$= \int_0^{\infty} e^{-x} x^{6-1} dx$$

$$= \Gamma 6$$

• Properties:

① $\Gamma n+1 = n \Gamma n = n!$

② $\Gamma \frac{1}{2} = \sqrt{\pi}$

③ $\Gamma 1 = 1$

④ $\Gamma m \Gamma 1-m = \frac{\pi}{\sin m\pi}$

The Beta function is defined by definite integral of the form

$$\beta(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$m, n > 0$

$$\longrightarrow \int_0^1 x^5 (1-x)^6 dx$$

$$= \int_0^1 x^{6-1} (1-x)^{7-1} dx$$

$$= \beta(6, 7)$$

• Properties:

① $\beta(m,n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$

* ② $\int_0^{\pi/2} \sin^m \theta \cos^n \theta \cdot d\theta = \frac{1}{2} \beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = \frac{1}{2} \frac{\Gamma \frac{m+1}{2} \Gamma \frac{n+1}{2}}{\Gamma \frac{m+n+2}{2}}$

Ex-1

Find the value of this integral $\int_0^{\pi/2} \sin^3 \theta \cos^4 \theta$ & convert this to integral in Beta Function.

Solⁿ: $= \frac{1}{2} B\left(\frac{3+1}{2}, \frac{4+1}{2}\right)$

$$= \frac{1}{2} B\left(2, \frac{5}{2}\right)$$

$$= \frac{1}{2} \cdot \frac{\Gamma 2 \Gamma \frac{5}{2}}{\Gamma 2 + \frac{5}{2}}$$

$$= \frac{\Gamma 2 \cdot \Gamma \frac{5}{2}}{2 \cdot \Gamma \frac{9}{2}}$$

$$= \frac{1! \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}}{2 \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}}$$

$$= \frac{2}{35} \text{ (Ans.)}$$

$$\cdot \Gamma_{n+1} = n! = n \Gamma n$$

$$\cdot \Gamma \frac{3}{2} = \Gamma \frac{1}{2} + 1 = \frac{1}{2} \Gamma \frac{1}{2}$$

$$\cdot \Gamma \frac{5}{2} = \Gamma \frac{3}{2} + 1 = \frac{3}{2} \Gamma \frac{3}{2} = \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}$$

$$\cdot \Gamma \frac{7}{2} = \Gamma \frac{5}{2} + 1 = \frac{5}{2} \Gamma \frac{5}{2} = \frac{5}{2} \cdot \frac{3}{2} \Gamma \frac{3}{2} = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}$$

Method 2

$$\Gamma \frac{5}{2} = \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}$$

$$\Gamma \frac{9}{2} = \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}$$

$$\Gamma \frac{7}{2} = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}$$

$$\frac{1!}{2 \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma \frac{1}{2}}$$

Ex-2:

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cdot \cos^n \theta \cdot d\theta$$

$$m=6$$

$$n=0$$

Comparing it with

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cos^n \theta \cdot d\theta$$

we obtain,

Solⁿ:

$$= \frac{1}{2} \beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$$

$$\left(\frac{1+1}{2}, \frac{1+0}{2}\right) \text{ where } m=6, n=0$$

$$= \frac{1}{2} \beta\left(\frac{7}{2}, \frac{1}{2}\right)$$

$$\left(\frac{7}{2}, \frac{1}{2}\right) \text{ where } m=6, n=0$$

$$= \frac{1}{2} \cdot \frac{\sqrt{\frac{7}{2}} \sqrt{\frac{1}{2}}}{\sqrt{\frac{7}{2} + \frac{1}{2}}}$$

$$\frac{\sqrt{\frac{7}{2}} \sqrt{\frac{1}{2}}}{\sqrt{\frac{7}{2} + \frac{1}{2}}} \cdot \frac{1}{2} =$$

$$= \frac{1}{2} \cdot \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{\sqrt{4}}$$

$$\frac{\sqrt{\frac{7}{2}} \sqrt{\frac{1}{2}}}{\sqrt{\frac{7}{2} + \frac{1}{2}}} =$$

$$= \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{\frac{1}{2} \cdot 2 \cdot 3! \cdot \frac{1}{2} \cdot \frac{1}{2}} = \frac{15}{32}$$

$$\frac{\sqrt{\frac{7}{2}} \sqrt{\frac{1}{2}}}{\sqrt{\frac{7}{2} + \frac{1}{2}}} =$$

$$= \frac{5\pi}{32} \sqrt{\pi} \cdot \sqrt{\pi}$$

$$\frac{15}{32} \cdot \frac{\pi}{2} =$$

$$= \frac{5\pi}{32}$$

Ex-3: $\int_0^{\pi/6} \sin^2 \theta \cos^3 3\theta \, d\theta$

Sol:
 $= \int_0^{\pi/6} (\sin 2z)^2 (\cos z)^2 \cdot \frac{dz}{3}$

$$= \frac{1}{3} \int_0^{\pi/6} (2 \sin z \cos z)^2 (\cos z)^2 \cdot dz$$

$$= \frac{4}{3} \int_0^{\pi/6} \sin^2 z \cos^4 z \cdot dz$$

$$= \frac{4}{3} \cdot \frac{1}{2} B\left(\frac{2+1}{2}, \frac{4+1}{2}\right)$$

$$= \frac{4}{3} \cdot \frac{1}{2} B\left(\frac{3}{2}, \frac{5}{2}\right) \longrightarrow (\text{Ans:}) \text{ if } B \text{ form এ রাখতে বলে}$$

$$= \frac{2}{3} \frac{\Gamma^{3/2} \Gamma^{5/2}}{\Gamma^{3/2+5/2}}$$

$$= \frac{2}{3} \cdot \frac{\frac{1}{2} \Gamma^{1/2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma^{1/2}}{\Gamma^4}$$

$$= \frac{2 \cdot \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma^{1/2} \Gamma^{1/2}}{3 \cdot 3!}$$

$$= \frac{\frac{3}{4} \sqrt{\pi} \cdot \sqrt{\pi}}{18}$$

$$= \frac{\pi}{24}$$

HW $\longrightarrow \int_0^{\pi/8} \sin^2 8\theta \cos^2 4\theta$

obj:

① $\sin(a) \sin(b)$

a $\rightarrow$ should be free without co-efficient

② $\int_0^{\pi/2}$ limit must be 0 to $\frac{\pi}{2}$

Let,

$z = 3\theta$ (Take smallest angle as z)

$$\Rightarrow dz = 3 \cdot d\theta$$

$$\Rightarrow d\theta = \frac{dz}{3}$$

θ	0	$\pi/6$
z	0	$3 \cdot \pi/6 = \pi/2$

(or)
 $\pi/6 \rightarrow \pi/2$
 $\Rightarrow \frac{\pi}{6} \times 3 = \frac{\pi}{2}$
 $\therefore z = 3\theta$

Ex-4

$$\int_0^{\infty} e^{-x^2} dx$$

$$= \int_0^{\infty} e^{-u} \cdot \frac{du}{2x}$$

$$= \frac{1}{2} \int_0^{\infty} e^{-u} \cdot \frac{du}{\sqrt{u}}$$

$$= \frac{1}{2} \int_0^{\infty} e^{-u} \cdot u^{-1/2} du$$

$$= \frac{1}{2} \int_0^{\infty} e^{-u} u^{1/2-1} du$$

$$= \frac{1}{2} \Gamma\left(\frac{1}{2}\right)$$

$$= \frac{1}{2} \cdot \sqrt{\pi}$$

$$= \frac{\sqrt{\pi}}{2}$$

Let,

$$x^2 = u \Rightarrow x = \sqrt{u}$$

$$\Rightarrow 2x \cdot dx = du$$

$$\Rightarrow dx = \frac{du}{2x}$$

$$\int_0^{\infty} e^{-x^2} x^{n-1} dx$$

$$\begin{aligned} &= \int_0^{\infty} e^{-u} \cdot \frac{du}{2} \cdot u^{n/2-1} \\ &= \frac{1}{2} \int_0^{\infty} e^{-u} u^{n/2-1} du \\ &= \frac{1}{2} \Gamma\left(\frac{n}{2}\right) \end{aligned}$$

$$\int_0^{\pi/2} \sin^{2n} \theta \cos^{2m} \theta d\theta$$

Ex-5 $\int_0^1 x^6 (1-x^2)^{1/2} dx$

$= \int_0^1 u^3 (1-u)^{1/2} \frac{du}{2 \cdot x}$

$= \int_0^1 u^3 (1-u)^{1/2} \frac{du}{2 \cdot \sqrt{u}}$

$= \frac{1}{2} \int_0^1 u^{3-\frac{1}{2}} (1-u)^{1/2} du$

$= \frac{1}{2} \int_0^1 u^{\frac{5}{2}} (1-u)^{1/2} du$

$= \frac{1}{2} \int_0^1 u^{\frac{7}{2}-1} (1-u)^{\frac{3}{2}-1} du$

$= \frac{1}{2} B\left(\frac{7}{2}, \frac{3}{2}\right) \rightarrow \text{Ans:}$

$= \frac{1}{2} \cdot \frac{\Gamma\left(\frac{7}{2}\right) \Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{7}{2} + \frac{3}{2}\right)}$

$= \frac{1}{2} \cdot \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}}{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}}$

$= \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}}}{2 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$

$= \frac{\frac{15}{16} \sqrt{\pi} \cdot \sqrt{\pi}}{2 \cdot 24}$

$= \frac{5\pi}{2 \cdot 128}$

$= \frac{5\pi}{256}$

Let,

$u = x^2 \Rightarrow x = \sqrt{u}$

$\Rightarrow du = 2x \cdot dx$

$\Rightarrow dx = \frac{du}{2x}$

$\Rightarrow dx = \frac{du}{2\sqrt{u}}$

Ex-6 $\int_0^1 x^6 (1-x^2)^{1/2} dx$

$= \int_0^1 u^3 (1-u)^{1/2} \frac{du}{2 \cdot x}$

$= \int_0^1 u^3 (1-u)^{1/2} \frac{du}{2 \cdot \sqrt{u}}$

$= \frac{1}{2} \int_0^1 u^{3-\frac{1}{2}} (1-u)^{1/2} du$

$= \frac{1}{2} \int_0^1 u^{\frac{5}{2}} (1-u)^{1/2} du$

$= \frac{1}{2} \int_0^1 u^{\frac{7}{2}-1} (1-u)^{\frac{3}{2}-1} du$

$= \frac{1}{2} B\left(\frac{7}{2}, \frac{3}{2}\right)$

$= \frac{1}{2} \cdot \frac{\Gamma\left(\frac{7}{2}\right) \Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{7}{2} + \frac{3}{2}\right)}$

$= \frac{1}{2} \cdot \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}}}{2 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$

$= \frac{15}{16} \sqrt{\pi} \cdot \sqrt{\pi}$

$= \frac{5\pi}{2 \cdot 128}$

$= \frac{5\pi}{256}$

Ex-6 $\int_0^1 \frac{dx}{(1-x^3)^{1/3}}$

$= \int_0^1 (1-u)^{-1/3} \cdot \frac{du}{3x^2}$

Let,
 $x^3 = u \Rightarrow x = u^{1/3} \Rightarrow x^2 = u^{2/3}$
 $3x^2 dx = du$
 $\Rightarrow dx = \frac{du}{3x^2}$

$= \frac{1}{3} \int_0^1 \frac{(1-u)^{-1/3}}{u^{2/3}} \cdot du$

$= \frac{1}{3} \int_0^1 u^{-2/3} (1-u)^{-1/3} \cdot du$

$= \frac{1}{3} \int_0^1 u^{1/3-1} (1-u)^{2/3-1} \cdot du$

$= \frac{1}{3} B\left(\frac{1}{3}, \frac{2}{3}\right)$

$= \frac{1}{3} \frac{\Gamma(1/3) \Gamma(2/3)}{\Gamma(1/3 + 2/3)}$

$= \frac{1}{3} \times \frac{\pi}{\sin \frac{\pi}{3}} \cdot \frac{1}{1!}$

$= \frac{\pi}{3 \sin \frac{\pi}{3}}$

$\frac{\Gamma(m) \Gamma(1-m)}{\Gamma(1)} = \frac{\pi}{\sin m\pi}$
 $\frac{\Gamma(1/3) \Gamma(2/3)}{\Gamma(1)} = \frac{\pi}{\sin \frac{\pi}{3}}$

HW

① $\int_0^1 z^4 (1-z)^3 dz ; B(5,4)$

② $\int_0^4 x^{3/2} (4-x)^{5/2} dx$

~~③ $\int_0^\infty x^{-1/2} e^{-x} dx$~~

④ $\int_0^\infty \frac{e^{-5x}}{\sqrt{x}} dx$

⑤ $\int_0^{\pi/2} \sqrt{1-\sin 2\theta} d\theta$

⑥ $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta$

⑦ $\int_0^a y^2 \sqrt{a^4 - y^4} dy$

⑧ $\int_0^\infty e^{-\sqrt{x}} \cdot x^{1/4} dx$

Ex-1: Proof that, $\Gamma_{1/2} = \sqrt{\pi}$

We know that, $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$

$$\therefore B(1/2, 1/2) = \int_0^1 x^{1/2-1} (1-x)^{1/2-1} dx$$

$$= \int_0^1 \frac{1}{\sqrt{x(1-x)}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{x-x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{-(x^2-x)}} dx$$

$$= \int_0^1 \frac{dx}{\sqrt{-\left[x^2 - 2 \cdot \frac{1}{2} \cdot x + \frac{1}{4} - \frac{1}{4}\right]}}$$

$$= \int_0^1 \frac{dx}{\sqrt{-\left[\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}\right]}}$$

$$= \int_0^1 \frac{dx}{\sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2}}$$

$$= \int_0^1 \frac{dx}{\sqrt{\left(\frac{1}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2}}$$

$$= \left[\sin^{-1} \left(x - \frac{1}{2} \right) \right]_0^1$$

$$= \sin^{-1}(1) - \sin^{-1}(-1) \quad [\because \sin^{-1}(-x) = -\sin^{-1}(x)]$$

$$= \frac{\pi}{2} + \sin^{-1}(1)$$

$$= \frac{\pi}{2} + \frac{\pi}{2}$$

$$= \pi$$

1	0	2
1	0	3

Now,

$$\beta(1/2, 1/2) = \pi$$

$$\Rightarrow \frac{\Gamma(1/2) \Gamma(1/2)}{\Gamma(1/2 + 1/2)} = \pi$$

$$\Rightarrow \frac{(\Gamma(1/2))^2}{\Gamma(1)} = \pi$$

$$\Rightarrow \Gamma(1/2) = \sqrt{\pi}$$

Hence $\Gamma(1/2) = \sqrt{\pi}$ (Proved)

(Note) $\Gamma(3/2) = \frac{\pi}{2}$

Ex(1) $\int_0^a y^z \sqrt{a^4 - y^4} dy$; Find the value, convert to Beta Function

Solⁿ: $[x]^{n/2} = (x-1)^{n/2}$ $(1-x)^{n/2} = (1-x)^{n/2}$

Let,
 $y^4 = z \cdot a^4$
 $\Rightarrow y = z^{1/4} \cdot a$
 $\Rightarrow \frac{dy}{dz} = a \cdot \frac{1}{4} z^{-3/4}$

y	0	a
z	0	1

$$\begin{aligned} \therefore \int_0^a y^z \sqrt{a^4 - y^4} dy &= \int_0^1 (a \cdot z^{1/4})^z \sqrt{a^4 - z \cdot a^4} \cdot \frac{a}{4} z^{-3/4} dz \\ &= \frac{a^z \cdot a^2 \cdot a}{4} \int_0^1 z^{z/4} (1-z)^{1/2} z^{-3/4} dz \\ &= \frac{a^{10}}{4} \int_0^1 z^{1/4} (1-z)^{1/2} dz \\ &= \frac{a^{10}}{4} \int_0^1 z^{3/4-1} (1-z)^{3/2-1} dz \\ &= \frac{a^{10}}{4} B(2, 3/2) \quad (\text{Ans:}) \end{aligned}$$

HW

Ex(3): $\int_0^3 y^2 \sqrt{16-y^4} dy$

Let $y^4 = 16z$

Ex(4): $\int_0^3 y^7 (\sqrt{9-y^2}) dy$; $\frac{34992}{35}$

Let $y^2 = 9z$

b3N * *

Integral on A #

$\int_{\Delta} f_{\Delta} + \int_{\Delta} f_{\Delta} = \int_{\Delta} f_{\Delta}$

$(\frac{1}{2}) \cdot 512 \cdot \beta(\frac{3}{4}, \frac{3}{2})$ Not sure $\int_{\Delta} f_{\Delta} < =$

$\int_{\Delta} (\frac{f_{\Delta}}{x_{\Delta}}) + 1 = \int_{\Delta} f_{\Delta} < =$

$\int_{\Delta} (\frac{f_{\Delta}}{x_{\Delta}}) + 1 \sqrt{\cdot} = \frac{f_{\Delta}}{x_{\Delta}} < =$

$\int_{\Delta} (\frac{f_{\Delta}}{x_{\Delta}}) + 1 \sqrt{\cdot} = \frac{f_{\Delta}}{x_{\Delta}} \sqrt{\cdot} < =$

$\int_{\Delta} (\frac{f_{\Delta}}{x_{\Delta}}) + 1 \sqrt{\cdot} = \frac{f_{\Delta}}{x_{\Delta}} < =$

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$\int_{\Delta} (\frac{f_{\Delta}}{x_{\Delta}}) + 1 \sqrt{\cdot} = \frac{f_{\Delta}}{x_{\Delta}} < =$

ok to ignore this

ok to ignore this

*** Mid

Arc Length:

$$(\Delta L)^2 = \Delta x^2 + \Delta y^2$$

$$\Rightarrow (\Delta L)^2 = \Delta x^2 \left[1 + \left(\frac{\Delta y}{\Delta x} \right)^2 \right]$$

$$\Rightarrow \left(\frac{\Delta L}{\Delta x} \right)^2 = 1 + \left(\frac{\Delta y}{\Delta x} \right)^2$$

$$\Rightarrow \frac{\Delta L}{\Delta x} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x} \right)^2}$$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \frac{\Delta L}{\Delta x} = \lim_{\Delta x \rightarrow 0} \sqrt{1 + \left(\frac{\Delta y}{\Delta x} \right)^2}$$

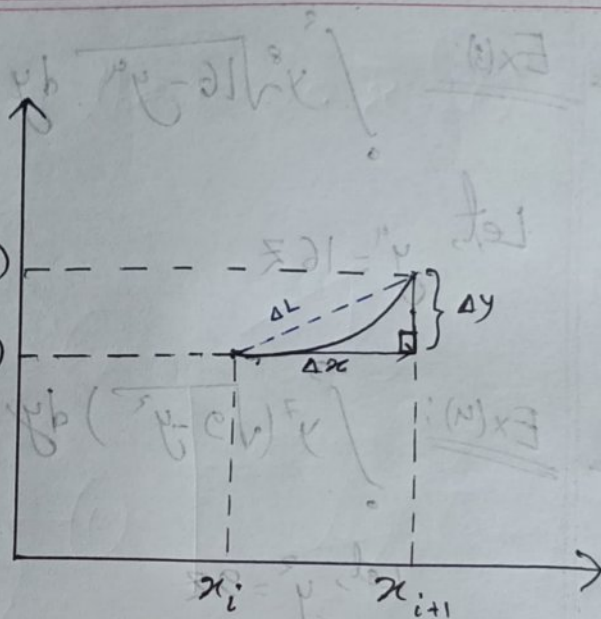
$$\Rightarrow \frac{dL}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

$$\Rightarrow \int_a^b dL = \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

$$\Rightarrow L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} \cdot dx$$

With respect to x

' x '



$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy} \right)^2} \cdot dy$$

with respect to y

' y '

Ex-1

Find the arc length of the curve $y = x^{3/2}$ from $(1, 1)$ to $(2, 2\sqrt{2})$ -

- i. with respect to x
- ii. with respect to y

Solⁿ:

(i) $L = \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$ — (i)

Now,

$$y = x^{3/2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3}{2} x^{1/2}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{9}{4} x$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{9x}{4} = \frac{9x+4}{4} \quad \text{--- (ii)}$$

(ii) = (i)

$$L = \int_1^2 \sqrt{\frac{9x+4}{4}} dx = \frac{1}{2} \int_1^2 \sqrt{9x+4} dx$$

$$= \frac{1}{2} \cdot \frac{1}{9} \int_{13}^{22} u \cdot du$$

$$= \frac{1}{18} \cdot \left(\frac{315}{2} \right)$$

$$= 8.75$$

Let,

$$9x+4 = u$$

$$\Rightarrow 9 \cdot dx = du$$

$$\Rightarrow dx = \frac{du}{9}$$

x	1	2
u	13	22

Ex-2

Find the arc length of the curve $x = \frac{1}{3}(y^2+2)^{3/2}$ from $y=0$ to $y=1$.

That means, with respect to y

Solⁿ:

$$L = \int_0^1 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \cdot dy$$

Finally,

$$x = \frac{1}{3}(y^2+2)^{3/2}$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{3} \cdot \frac{3}{2} (y^2+2)^{1/2} \cdot 2y = y(y^2+2)^{1/2}$$

$$\Rightarrow \left(\frac{dx}{dy}\right)^2 = y^2(y^2+2)$$

$$\Rightarrow 1 + \left(\frac{dx}{dy}\right)^2 = 1 + y^4 + 2y^2 = (y^2)^2 + 2 \cdot y^2 \cdot 1 + (1)^2 = (y^2+1)^2$$

$$\therefore \textcircled{ii} \Rightarrow \textcircled{i}:$$

$$L = \int_0^1 \sqrt{(1+y^2)^2} \cdot dy = \int_0^1 (1+y^2) \cdot dy = \left[y + \frac{y^3}{3} \right]_0^1 = 1 + \frac{1}{3} = \frac{4}{3}$$

5	2	18
55	01	10

$$\left(\frac{218}{5}\right) \cdot \frac{1}{5} = \frac{14}{83} =$$

$$25.8 =$$

Parametric Eqⁿ → Arc Length:

$$x = x(t)$$

$$y = y(t), \quad a \leq t \leq b$$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \cdot dt$$

Ex-1:

Find the arc length of the parametric curve -

$$x = \cos(2t)$$

$$y = \sin(2t) ; 0 \leq t \leq \frac{\pi}{2}$$

$$\frac{dx}{dt} = -2\sin(2t)$$

$$\frac{dy}{dt} = 2\cos(2t)$$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \cdot dt$$

$$= \int_0^{\pi/2} \sqrt{(-2\sin(2t))^2 + (2\cos(2t))^2} \cdot dt$$

$$= 2 \int_0^{\pi/2} dt$$

$$= 2 [t]_0^{\pi/2}$$

$$= 2 \cdot \frac{\pi}{2}$$

$$= \pi$$

HW

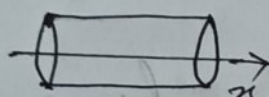
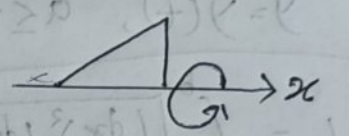
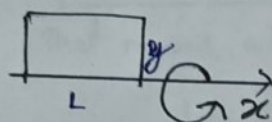
Find the arc length of the parametric curve: $x = \cos t + t \cdot \sin t$
 $y = \sin t - t \cos t$

$$0 \leq t \leq \pi$$

~~HW~~ → Find the arc length of the parametric curve.

$$x =$$

*** Mid *** # Surface of Revolution: - Bend



Cylinder



Sphere



$$A = \text{Length} \times \text{Width}$$

$$= L \times 2\pi y$$

$$dA = 2\pi y L$$

$$\Rightarrow dA = \int 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow A = \int 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

'x'

$$A = \int 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

'y'

Ex: Find the area of the surface that is generated by the portion of the curve $y=x^2$ between $x=1$ to $x=2$ about y -axis.

Solⁿ: $\rightarrow$ mainly, y -axis er respect e ber kante hbe...

$$y=x^2$$

x	1	2
y	1	4

$$\therefore A = \int_1^4 2\pi x \cdot \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \cdot dy$$

$$= \int_1^4 2\pi \sqrt{y} \cdot \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \cdot dy$$

$$= \int_1^4 2\pi \sqrt{y} \cdot \sqrt{\frac{4y+1}{4y}} \cdot dy$$

$$= \int_1^4 2\pi \cdot \sqrt{y} \cdot \frac{1}{2\sqrt{y}} \sqrt{4y+1} \cdot dy$$

$$= \pi \int_1^4 \sqrt{4y+1} \cdot dy$$

$$= \pi \int_5^{17} \sqrt{u} \cdot du$$

$$= \pi \cdot 30.275 \times \pi$$

~~30.275~~

$$x = \sqrt{y}$$

$$\frac{dx}{dy} = \frac{1}{2\sqrt{y}}$$

$$\left(\frac{dx}{dy}\right)^2 = \frac{1}{4y}$$

$$1 + \left(\frac{dx}{dy}\right)^2 = \frac{4y+1}{4y}$$

Let,

$$4y+1 = u$$

$$\Rightarrow 4 \cdot dy = du$$

$$\Rightarrow dy = \frac{du}{4}$$

y	1	4
u	5	17